

Assessment

1. What is the difference between **Descriptive** and **Inferential Statistics**?
2. Define the following terms:
 - Population
 - Sample
3. Define **mean**, **median**, and **mode**. How are they different from each other?
4. What is a Random Variable? What are its types?

Practical

[https://github.com/Divyraj-](https://github.com/Divyraj-Tops/Excel/blob/main/Data/Student%20Attitude%20and%20Behavior.csv)

[Tops/Excel/blob/main/Data/Student%20Attitude%20and%20Behavior.csv](https://github.com/Divyraj-Tops/Excel/blob/main/Data/Student%20Attitude%20and%20Behavior.csv)

1. Use the “**Student Attitude and Behavior**” dataset and select the following columns: **Height, Weight, 10th Marks, 12th Marks**.
 - Find Mean, Median, Mode and Quartiles
 - Find Skewness, kurtosis and Interpretation Based on the Skewness, kurtosis.
2. An investment analyst wants to assess the relationship between the stock prices of two companies to identify potential investment opportunities. Data: Let's consider the daily closing prices (in dollars) of Company A and Company B for a sample of 20 trading days:
Company A: 45, 47, 48, 50, 52, 53, 55, 56, 58, 60, 62, 64, 65, 67, 69, 70, 72, 74, 76, 77
Company B: 52, 54, 55, 57, 59, 60, 61, 62, 64, 66, 67, 69, 71, 73, 74, 76, 78, 80, 82, 83
 - Calculate the covariance and correlation between the stock prices of Company A and Company B. Interpret the value of the covariance and explain the nature of the relationship between the two stocks.
3. You want to test if there is a significant difference between the means of these two groups (at the 5% significance level).
Group 1 scores: [85, 88, 92, 87, 90]
Group 2 scores: [78, 80, 75, 82, 76]